



Depth-Aware Portrait Effects

A Comparative Implementation: MiDaS vs. Depth-Anything

CHAPTER 19 | MOBILE COMPUTATIONAL PHOTOGRAPHY

CASE STUDY

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The Physical Gap: Why Mobile Needs Math to Mimic Optics

Optical Limitations

Glass & Physics



Large Sensor & Aperture (f/1.4).

Physical depth of field.

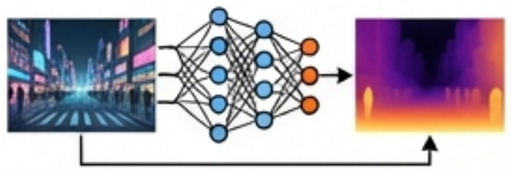
SENSOR SIZE: ~860mm²

LENS DIAMETER: >75mm

FOCAL LENGTH: 24-70mm

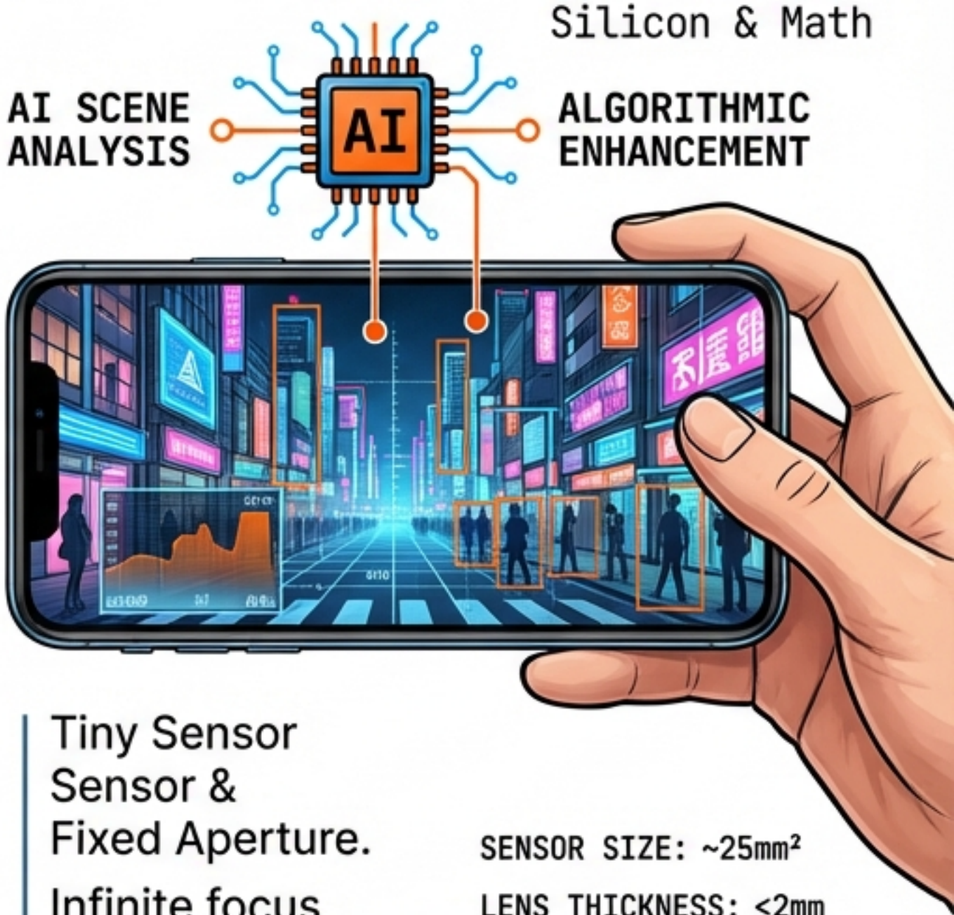
APERTURE: f/1.4 - f/22

Bridged by Monocular Depth Estimation



Algorithmic Enhancement

Silicon & Math



Tiny Sensor Sensor & Fixed Aperture.

Infinite focus requiring AI prediction.

SENSOR SIZE: ~25mm²

LENS THICKNESS: <2mm

FOCAL LENGTH: ~26mm (EFL)

APERTURE: f/1.8 (FIXED)

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Implementation Objectives: Defining Success



Depth Accuracy

Precise separation of foreground subject and background layers.



Face Protection

Semantic awareness to ensure facial features remain razor-sharp.



Real-Time Speed

Inference must occur in **< 1.0 seconds** for user flow.



Aesthetic Quality

Smooth, natural bokeh transitions without cutout artifacts.

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The Contenders: A Tale of Two Architectures

Contender 01: MiDaS

The Reliable Veteran

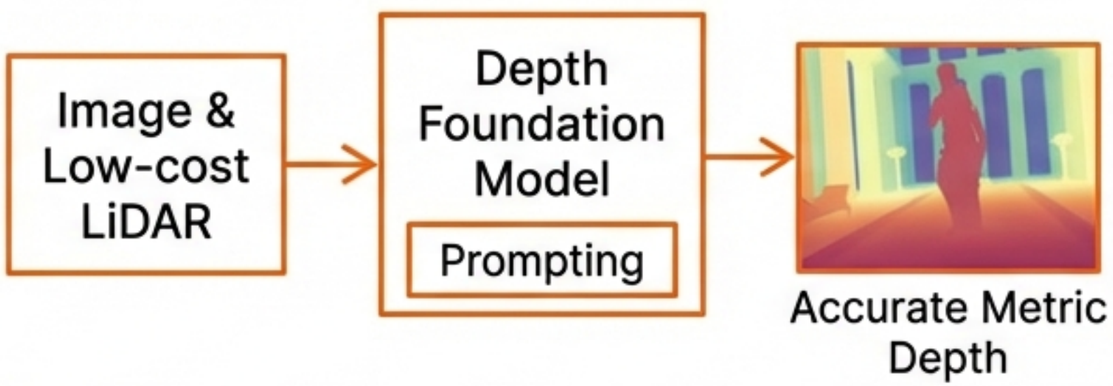
- CNN Architecture
- Mixed-data training
- Stable & Proven
- ~100MB Model Size

VS

Contender 02: Depth-Anything

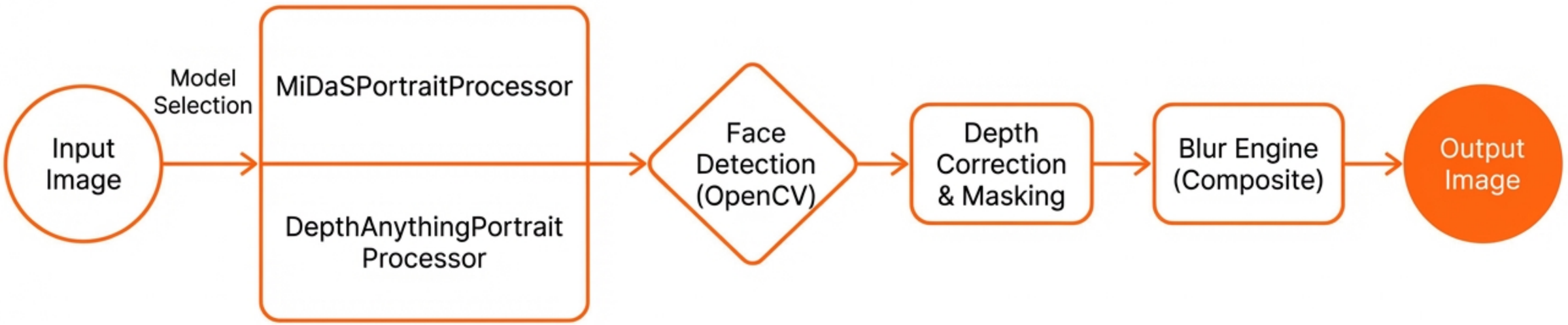
The Challenger

- Vision Transformer (DINOv2)
- Foundation Model
- Optimized for Speed
- ~200MB Model Size



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System Architecture: A Modular Pipeline

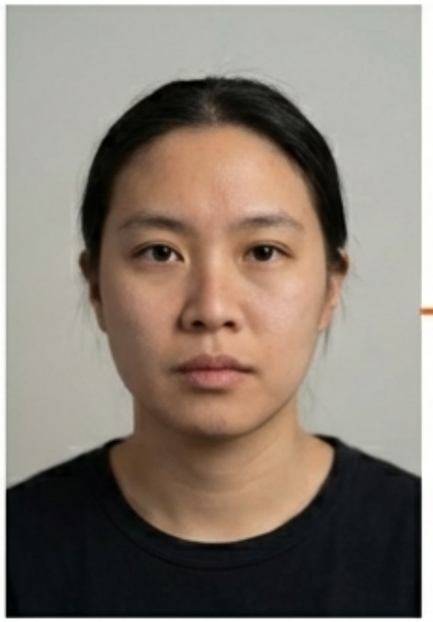



```
video_bokeh/
├── approaches/
│   ├── midas/
│   └── depth_anything/
└── comparison/
```

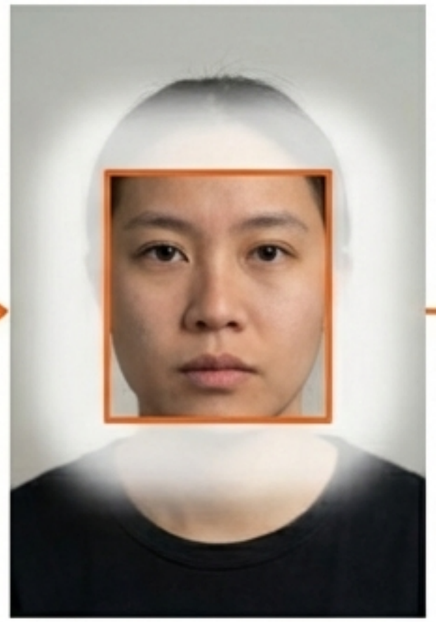
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The ‘Secret Sauce’: Hybrid Face Protection

Solving the ‘Blurred Face’ artifact using Classical CV



Input
Newsreader



Expanded
Soft Mask



Protected
Depth Map

```
def _detect_faces(self, image):
    # OpenCV Haar Cascade
    faces = self.face_cascade.detectMultiScale(gray)

    # Expand & Blur
    margin = 0.2
    face_region = cv2.GaussianBlur(face_region, (21, 21), 0)
```

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Corrected Depth Logic & Interpretation



Value 0.0
Background
Maximum Blur

Value 1.0
Foreground
Sharp Focus

1. **Normalization:** Scale raw depth to 0-1 range.
2. **Inversion:** Ensure 'Near' is High (1.0).
3. **Mask Overlay:** Apply Face Mask as 'Force Sharp' (1.0).
4. **Kernel Map:** Depth Value -> Blur Radius.

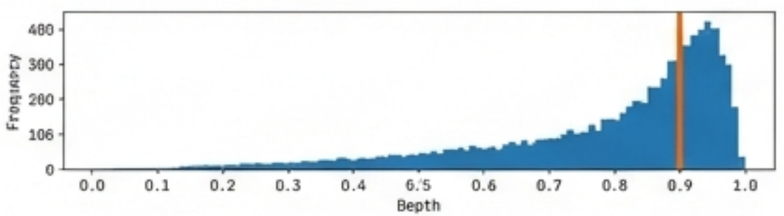
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Adaptive Processing Strategies

Conservative

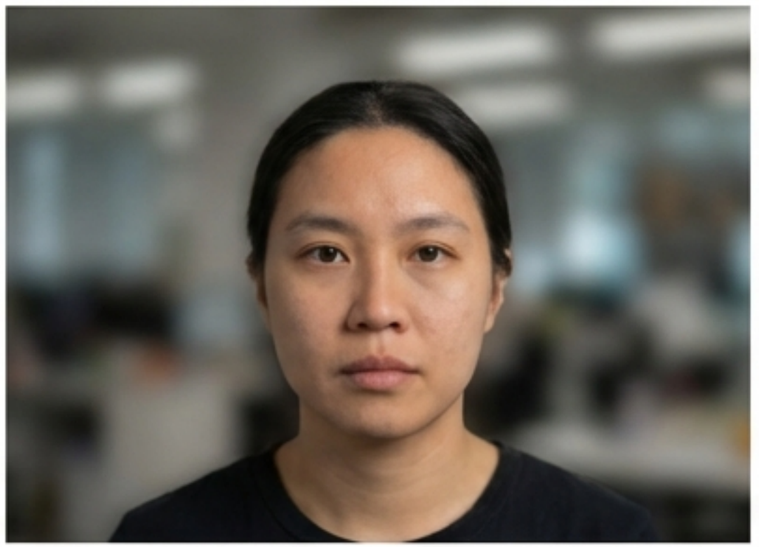


Threshold **0.8**

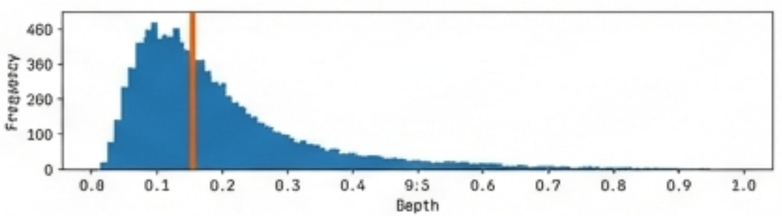


"Environmental Context"

Face-Aware

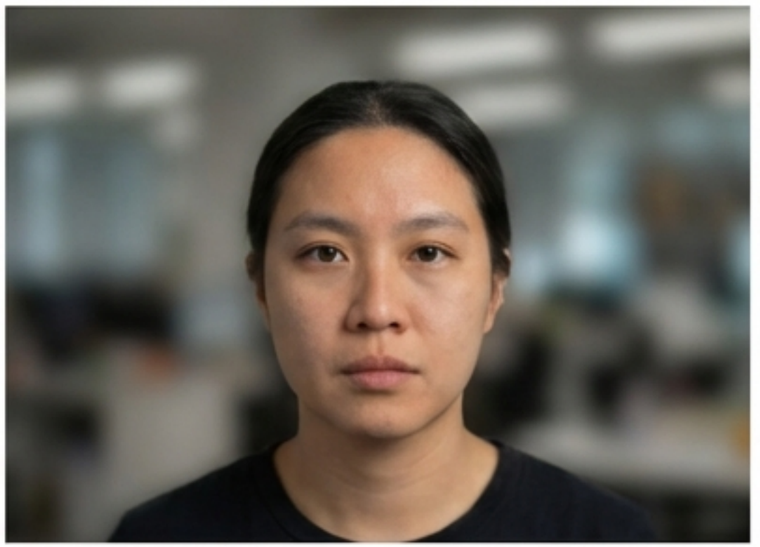


Threshold **0.2**

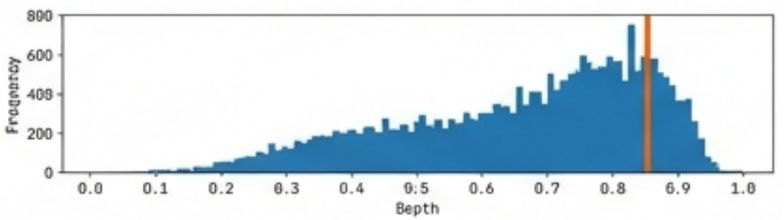


"Social Media Style"

Adaptive



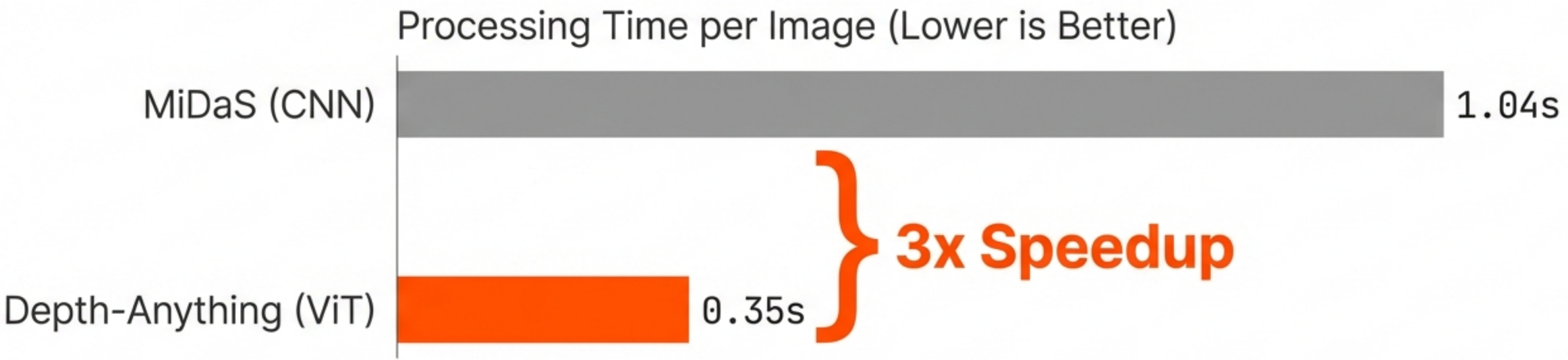
85th Percentile



"Dynamic Calculation"

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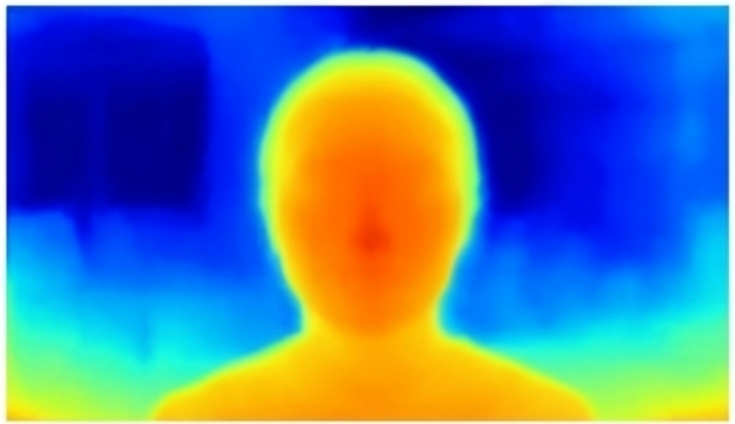
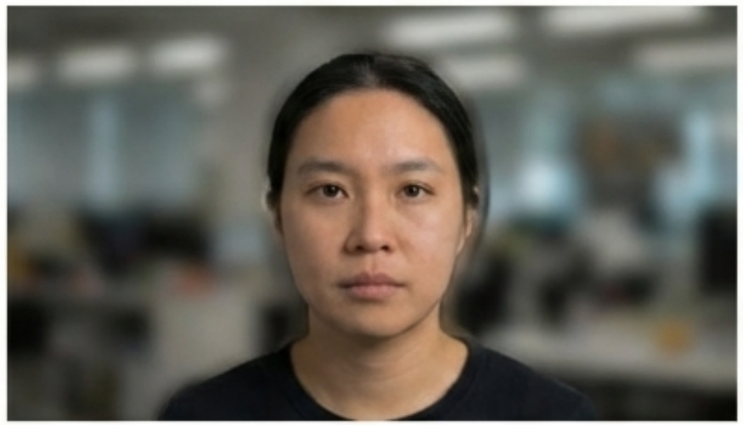
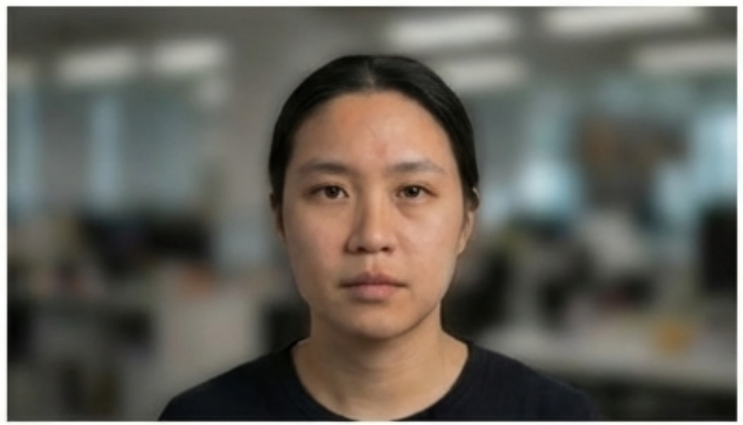
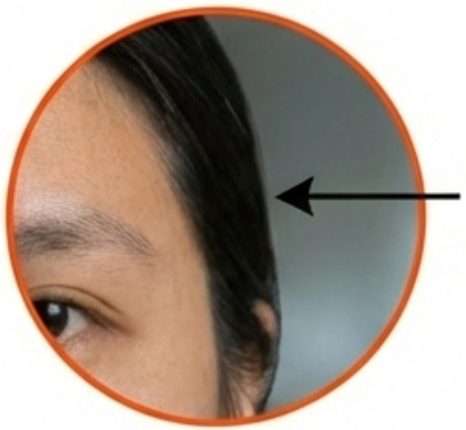
Performance Benchmarks: The Speed Breakthrough



Hardware Setup: Intel Core i7, CPU Inference. Threshold for mobile usability met at < 0.5s.

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Visual Quality Assessment

	Raw Depth Map	Final Composite	Detail Zoom
MiDaS			 Edge Bleed
Depth-Anything			 Clean Separation

Note in Newsreader: Face Protection Success Rate: 100% on both models.

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Key Technical Insights

Foundation Models Win

The DINOv2 backbone in Depth-Anything provides richer semantic understanding of scene structure than older CNNs.

Newsreader

Hybrid Robustness

AI alone is not enough. Classical Computer Vision (OpenCV Face Detection) provides the necessary safety layer for production apps.

Newsreader

Blur Realism

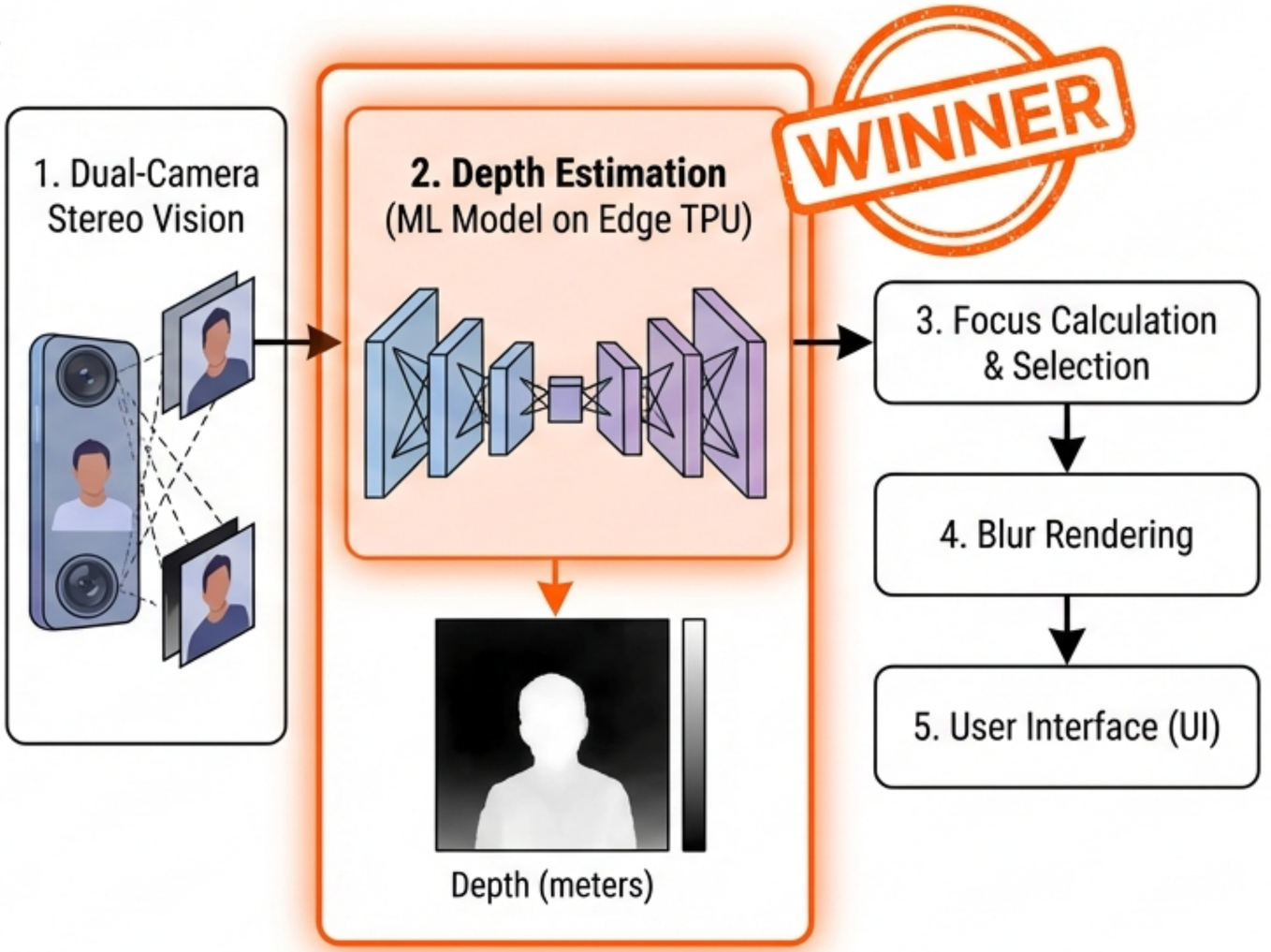
Simple Gaussian blur is insufficient. Realistic bokeh requires circular kernels and alpha blending to mimic optical physics.

Newsreader

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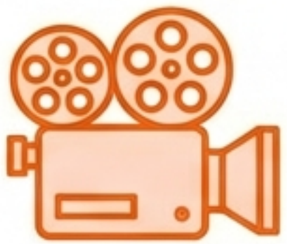
Why ‘Depth-Anything’ is the Mobile Choice

1. **Efficiency:** 0.35s latency preserves battery and user flow.
2. **Scalability:** Foundation model architecture improves with unlabeled data.
3. **Precision:** Solves scale inaccuracy for metric-like depth.



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Future Directions & Research Opportunities



Real-Time Video

GPU acceleration to reach **30fps** video bokeh.

Newsreader



Semantic Awareness

Expanding protection masks to pets, bodies, and objects.

Newsreader



On-Device Training

Fine-tuning models on user's specific photo style.

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Conclusion

We successfully implemented a modular portrait system comparing MiDaS and Depth-Anything. The verdict is clear: Depth-Anything represents a generational leap, offering superior **edge quality** at **3x the speed**.

Get the Code:
`github.com/JeffRen1977/video_bokeh`

Based on Chapter 19 | Mobile Computational Photography

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